Project Overview

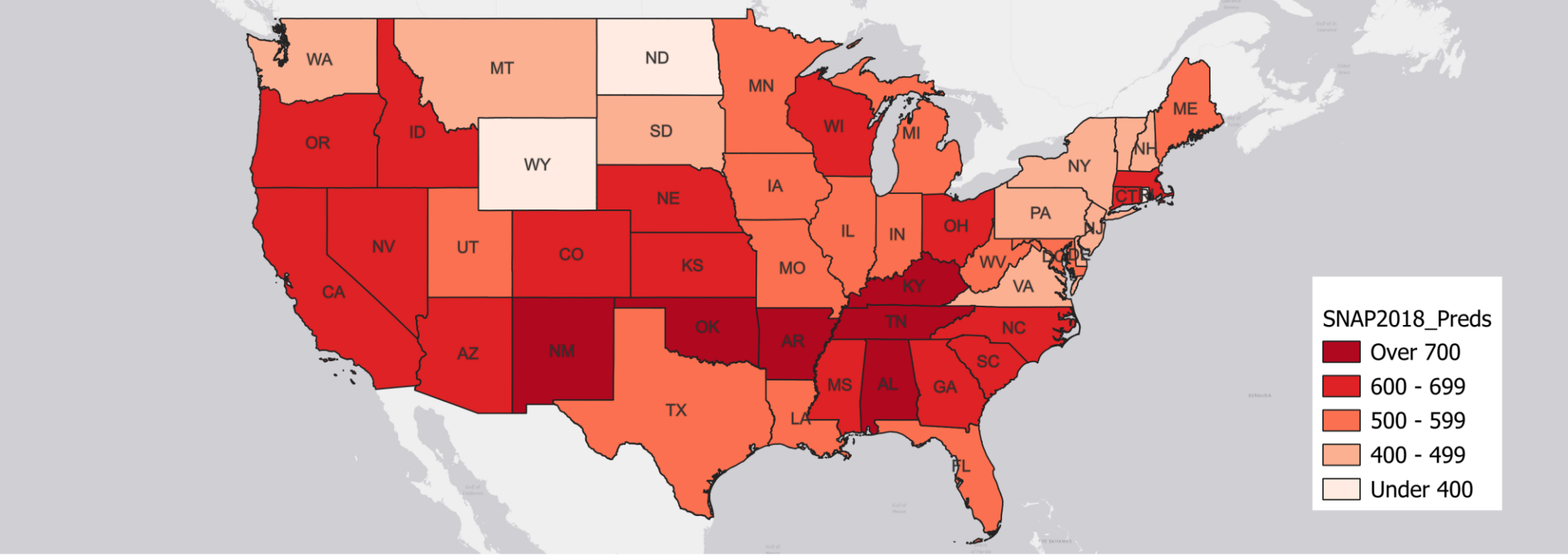
This project analyzes state-level Supplemental Nutrition Assistance Program (SNAP) prediction data to explore geographic patterns in projected food assistance demand across the United States.

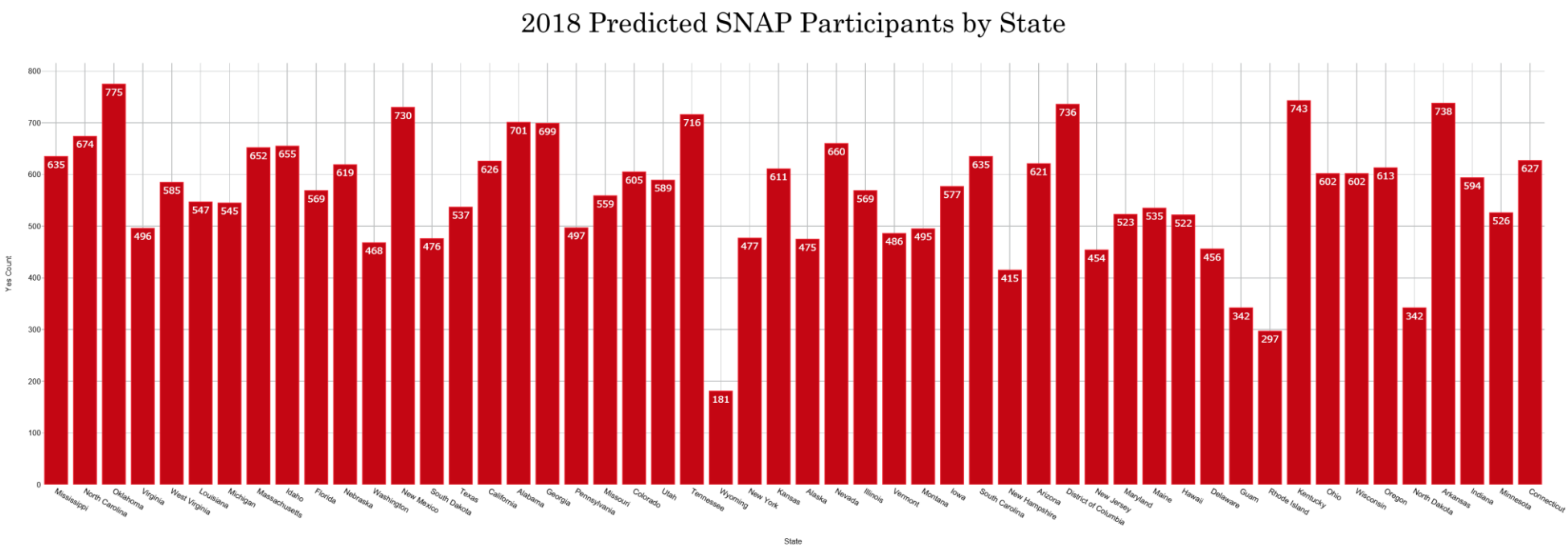
Using Python for data processing and QGIS for spatial visualization, the analysis maps three indicators related to SNAP participation and housing-related eligibility deductions.

By visualizing these variables spatially, the project highlights how housing costs and food assistance demand vary geographically across the United States.

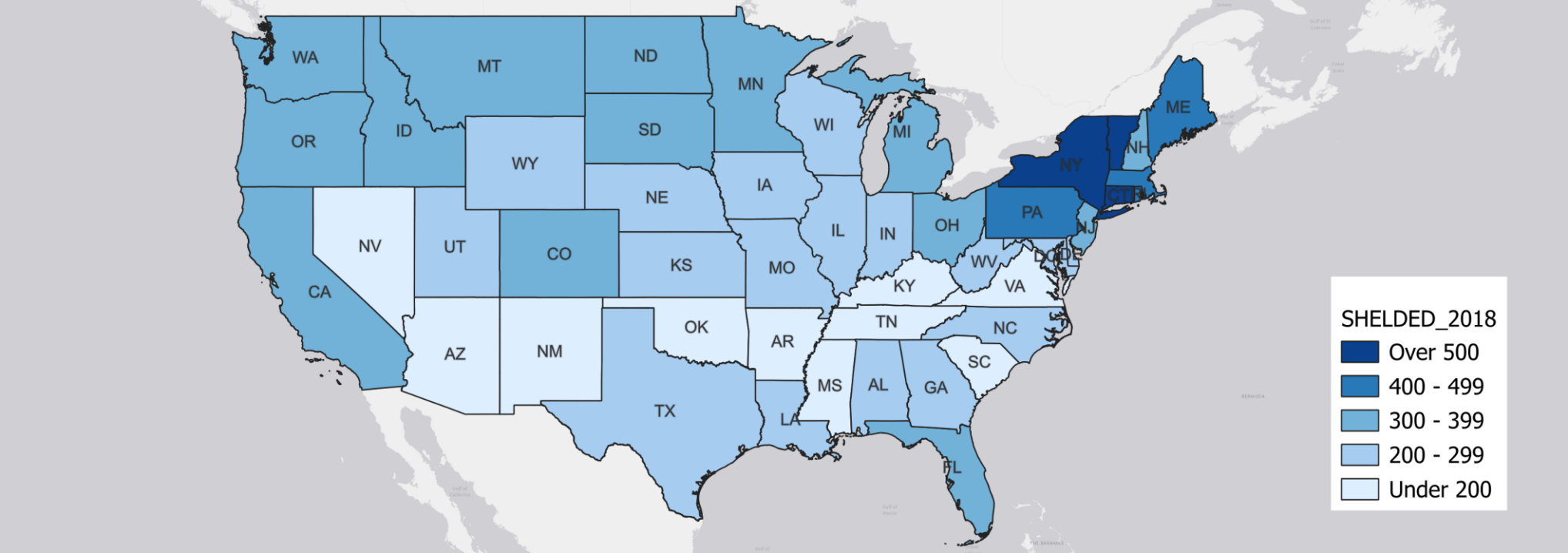
Variables Analyzed

- **Predicted SNAP Participants**
  Estimated number of individuals expected to receive SNAP benefits in each state based on modeled economic conditions.
- **Reported Shelter Deductions**
  Total housing-related deductions (rent, mortgage, and utilities) reported by SNAP households. Higher values indicate a greater share of household income being spent on housing.
- **Excess Shelter Deductions**
  Additional deductions applied when housing costs exceed 50% of a household's adjusted income. Higher values may indicate greater financial strain from housing costs.

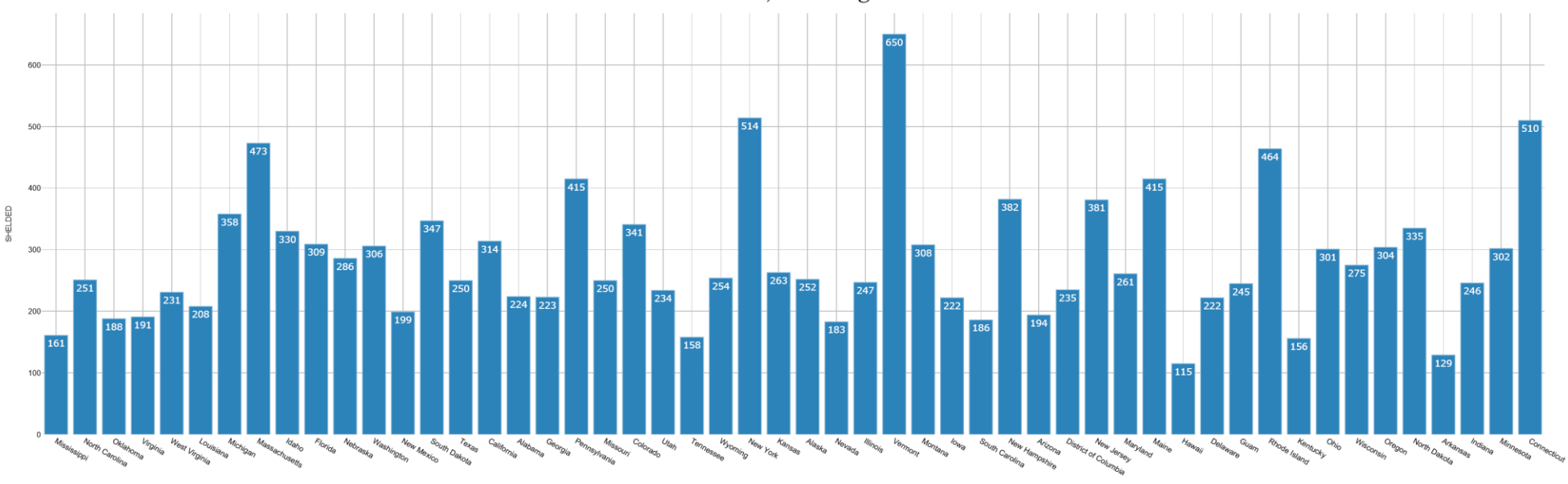




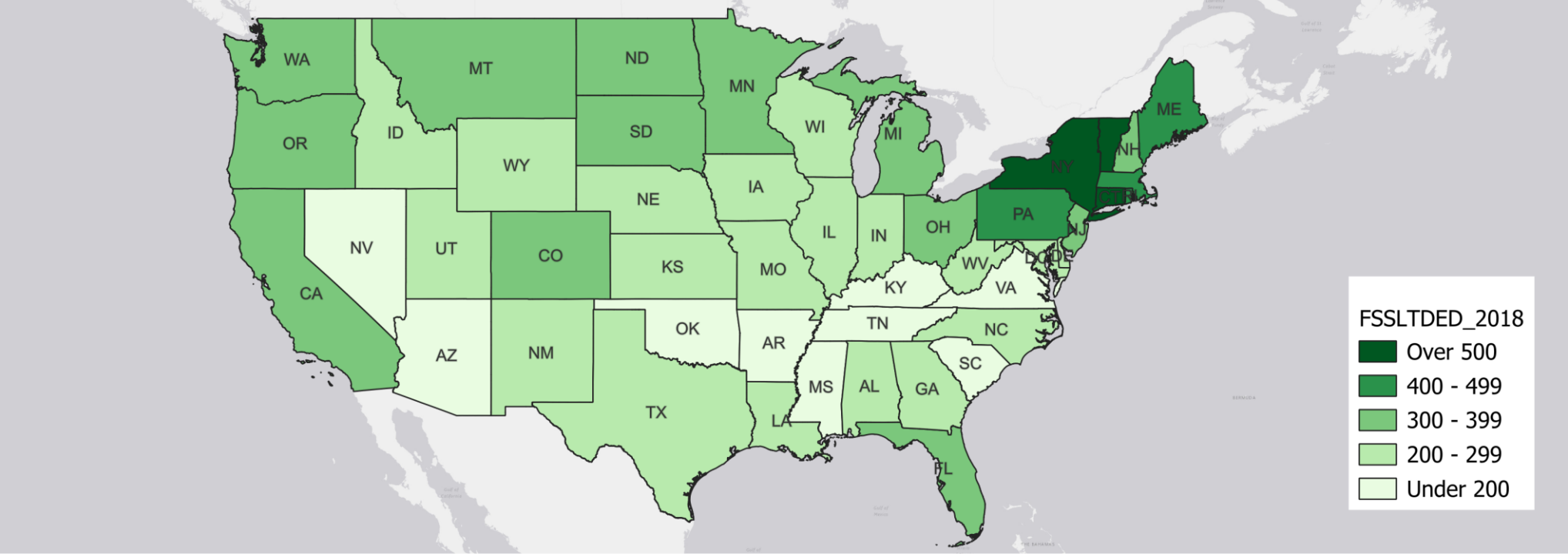
Predicted SNAP participation appears highest in several southeastern and southwestern states, suggesting stronger demand for food assistance in these regions relative to the national average.



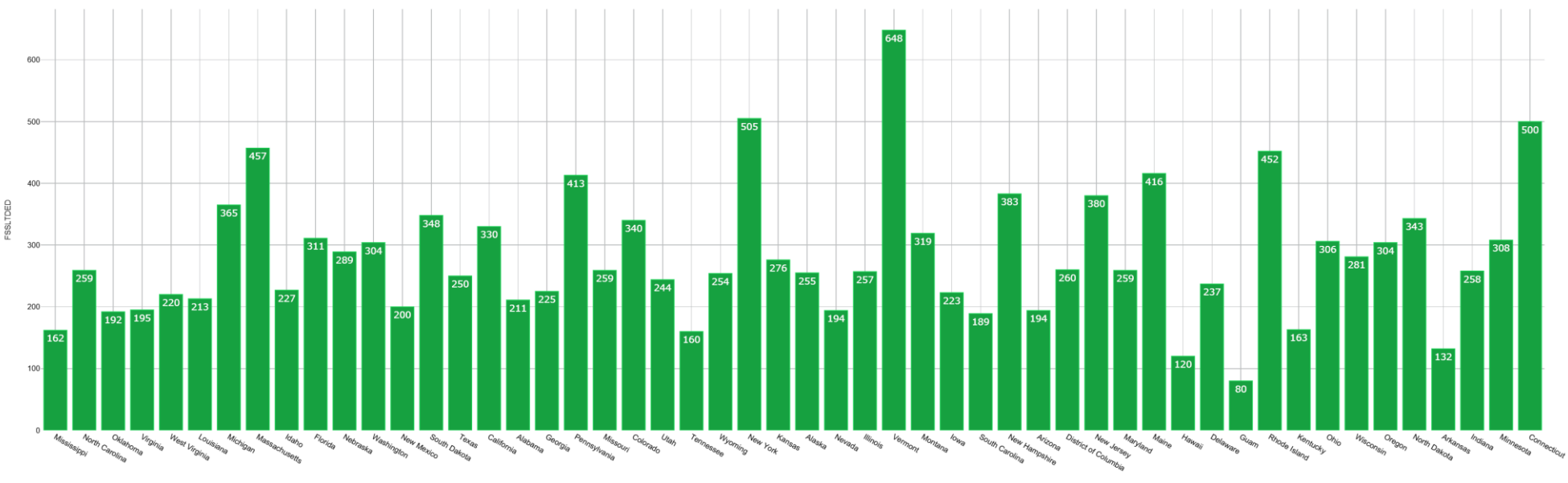
Reported Shelter Deduction: Average by State
(the taller the bar, the larger deduction allowed)



States with higher housing costs show larger reported shelter deductions, indicating that housing expenses play a larger role in SNAP eligibility calculations in these regions.



Calculated Excess Shelter Deduction: Averge by State
(the taller the bar, the larger deduction allowed)



Excess shelter deductions are highest in states where housing costs frequently exceed SNAP program thresholds, meaning housing expenses may have a greater impact on determining benefit eligibility.

Key Takeaways & Insights

SNAP participation patterns do not perfectly align with housing deduction patterns. Some regions with high SNAP participation show relatively lower housing deductions, suggesting that factors beyond housing costs—such as income levels and employment conditions—also play an important role in program participation.

Mapping SNAP-related variables reveals that both program participation and housing-related deductions vary significantly across states, reflecting differences in housing markets, income levels, and regional economic conditions.

Because the analysis uses state-level averages, it may mask important variation within states, particularly in large or economically diverse regions.